

Risk Classification Modeling Report

1. Executive Summary

The objective of this project was to build a robust machine learning pipeline for predicting user risk profiles ('RiskFlag'). To achieve high predictive accuracy while managing computational constraints, we implemented and evaluated three distinct architectures: a Deep Neural Network (MLP), a Support Vector Machine (SVM), and a Logistic Regression baseline.

By implementing advanced preprocessing techniques such as Quantile Transformation and specialized training strategies like Bagging and Learning Rate Scheduling, we achieved the following performance results:

Model Architecture	Accuracy
Deep Learning (MLP)	0.888
Logistic Regression	0.887
Bagged SVM	0.884

2. Preprocessing & Feature Engineering

A unified preprocessing pipeline was applied across all models to ensure consistency. Raw data often contains latent patterns that simple values cannot capture, so we applied specific transformations to expose these signals to the models.

2.1 Feature Engineering Strategies

- Row-Wise Statistical Aggregations: For every numeric feature, we calculated the Mean, Standard Deviation, Min, and Max across the row. This captures the volatility and range of user behavior rather than just static snapshots.
- Frequency Encoding: Applied to all categorical columns. This converts categories into numerical values based on their prevalence, allowing the model to weigh 'rare' versus 'common' profiles differently.

2.2 The Quantile Transformation Strategy

A critical decision in this pipeline was replacing the standard scaler with the QuantileTransformer. This non-linear transformation maps the input features to a Gaussian (Normal) Distribution. Deep Neural Networks and RBF-Kernel SVMs perform significantly better when input features follow a Bell Curve, leading to faster convergence and better margin separation.

3. Model Architectures

3.1 Deep Learning: GPU-Accelerated MLP

The top-performing model (0.888) was a custom Deep Neural Network built in PyTorch using a tapered 'funnel' architecture:

- Structure: Input -> Linear(512) -> Linear(256) -> Linear(128) -> Output.
- Activation: We utilized GELU (Gaussian Error Linear Unit) instead of ReLU for smoother gradient flow.
- Regularization: Heavy Dropout (0.35) and Batch Normalization were applied after every hidden layer to prevent overfitting.
- Scheduler: The 'OneCycleLR' scheduler was implemented. This warms up the learning rate to a peak (1e-2) to escape local minima and then decays it, ensuring a stable, generalized solution.

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3.2 Non-Linear Modeling: Bagged SVM

Training a standard SVM on this dataset size is typically computationally infeasible due to cubic complexity $O(N^3)$. To resolve this while retaining the accuracy of the RBF kernel, we implemented a Bagging Ensemble:

- Ensemble Strategy: Instead of one massive SVM, we used a BaggingClassifier to train 10 smaller, independent SVMs in parallel.
- Subsampling: Each estimator was trained on a random 15% subset of the data (`max_samples=0.15`).
- Result: This reduced training time from hours to minutes while maintaining non-linear decision capabilities, achieving an accuracy of 0.884.

4. Training & Validation Methodology

To ensure that our results were robust and not due to lucky data splits, we employed a Stratified 5-Fold Cross-Validation scheme. This ensures that the ratio of Risk Flags (0 vs 1) remains constant across all training and validation folds.

Furthermore, we did not assume a default decision threshold of 0.5. We performed a dynamic threshold search (from 0.2 to 0.8) on the Out-of-Fold (OOF) predictions to find the precise decision boundary that maximized accuracy for each model.

5. Conclusion

The analysis demonstrates that while the dataset has strong linear components (evidenced by the Logistic Regression's score), the Deep MLP architecture successfully extracted marginal gains through complex feature interactions and the OneCycle learning strategy. Additionally, the Bagged SVM approach proved that expensive non-linear kernels can be made viable for large datasets through ensemble subsampling.

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